NMEC595 – Research Methodology

Lecture: Chi-Square Tests & Introduction to ANOVA

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Where These Topics Fit

What We Have Already Seen

- Inference for one mean or one proportion
- Comparing two means or two proportions

What We Study Now

- **One categorical variable (multi-level)** $\rightarrow$ **Chi-Square Goodness-of-Fit**
- **Two categorical variables** $\rightarrow$ **Chi-Square Test for Independence**
- **One categorical** $+$ **one quantitative variable** $\rightarrow$ **ANOVA**
- **Two quantitative variables** $\rightarrow$ regression/correlation (later)

Intuition: The first question to ask is: what type of variables do I have? That determines which method to use.

A Goodness-of-Fit Test

A chi-square goodness-of-fit test checks whether observed counts match a claimed distribution.

Multinomial Experiment

An experiment is multinomial if:

1. it has n identical trials,
2. each trial falls into one of k categories, with $k > 2$,
3. the trials are independent,
4. the category probabilities stay constant.

Intuition: We compare the sample counts with the counts expected under H_0.

Observed and Expected Frequencies

Definitions

Observed frequencies (O): the actual counts from the sample.
Expected frequencies (E): the counts expected if H_0 is true.

Expected Count Formula

$$E = np$$

where n is the sample size and p is the hypothesized category probability.

Chi-Square Test Statistic (Goodness-of-Fit)

Formula

$$\chi^{2*} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

Degrees of Freedom

$$df = k - 1$$

- Large differences between O and E give a large χ^2
- The test is always **right-tailed**
- Each expected count should be **at least 5**

Decision Rule (Goodness-of-Fit)

Critical Value Method

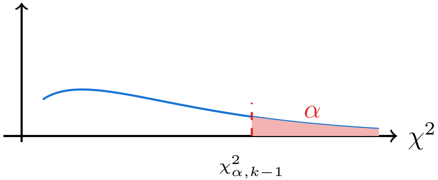
Reject H_0 if

$$\chi^{2*} > \chi^2_{\alpha,\,(k-1)}$$

p-Value Method

Reject H_0 if

$$p\text{-value} < \alpha$$



Example 1: Candy Color Distribution

A candy company claims: 30% Red, 20% Blue, 25% Green, 25% Yellow. A sample of $n = 200$ candies is drawn. Compute expected counts.

Color	**Claimed** p	**Expected** $E = np$	**Observed** O
Red	0.30	$200(0.30) = 60$	48
Blue	0.20	$200(0.20) = 40$	52
Green	0.25	$200(0.25) = 50$	41
Yellow	0.25	$200(0.25) = 50$	59
Total		200	200

Intuition: Under H_0, each expected count equals sample size $\times$ claimed probability.

Example 1: Computing the Statistic

Color	Observed Count	Expected Count
Red	48 (60)	60
Blue	52 (40)	40
Green	41 (50)	50
Yellow	59 (50)	50
Total	200	200

$$\chi^{2*} = \frac{(48-60)^2}{60} + \frac{(52-40)^2}{40} + \frac{(41-50)^2}{50} + \frac{(59-50)^2}{50}$$

$$= \frac{144}{60} + \frac{144}{40} + \frac{81}{50} + \frac{81}{50} = 2.40 + 3.60 + 1.62 + 1.62 = \mathbf{9.24}$$

$$df = k - 1 = 4 - 1 = 3, \qquad p\text{-value} \approx 0.026$$

Example 1: Conclusion

- Test statistic: $\chi^{2*} = 9.24$
- Degrees of freedom: $df = 3$
- Critical value at $\alpha = 0.05$: $\chi^2_{0.05,3} = 7.815$

Decision

Since $9.24 > 7.815$, we **reject** H_0.

Conclusion in Context

There is sufficient evidence that the actual color distribution differs from the company's claimed distribution.

Chi-Square Test for Independence: Overview

What Does It Test?

The Chi-Square test for independence determines whether **two categorical variables are associated** or independent of each other.

Contingency Tables

Definition

A **contingency table** (two-way table) summarises counts for two categorical variables simultaneously.

Table Size

- r = number of row categories
- c = number of column categories
- Table size $= r \times c$ (margin totals are **not** counted)

Intuition: Each cell counts observations that belong to *one* row category *and* one column category simultaneously.

Chi-Square Hypotheses

Null Hypothesis

$$H_0: \text{ The two categorical variables are } \textbf{independent} \text{ (no association)}$$

Alternative Hypothesis

$$H_a: \text{ The two categorical variables are } \textbf{dependent} \text{ (associated)}$$

Equivalent Language

"Independent" $\Leftrightarrow$ **no association**. "Dependent" $\Leftrightarrow$ **there is an association**.

Intuition: We assume no relationship first, then ask whether the data deviate too far from that assumption.

Exercise Problem

Violence and lack of discipline have become major problems in schools in the United States. A random sample of 300 adults was selected, and these adults were asked if they favor giving more freedom to schoolteachers to punish students for violence and lack of discipline. The two-way classification of the responses of these adults is presented in the following table.

	In Favor **(*F*)**	**Against** **(*A*)**	**No Opinion** **(*N*)**
Men (*M*)	93	70	12
Women (*W*)	87	32	6

Calculate the expected frequencies for this table, assuming that the two attributes, gender and opinions on the issue, are independent.

Expected Counts Under Independence

Expected Count Formula

$$E = \frac{(\text{row total}) \times (\text{column total})}{\text{grand total}}$$

Why This Formula?

Under independence, the joint probability equals the product of the marginal probabilities:

$$P(\text{row } i \cap \text{col } j) = P(\text{row } i) \cdot P(\text{col } j)$$

Chi-Square Test Statistic (Independence)

Formula

$$\chi^{2*} = \sum_{\text{all cells}} \frac{(O - E)^2}{E}$$

Degrees of Freedom

$$df = (r - 1)(c - 1)$$

- Large discrepancies between O and E produce a large χ^{2*}
- The test is always **upper-tailed**

Decision Rule (Independence Test)

Critical Value Method

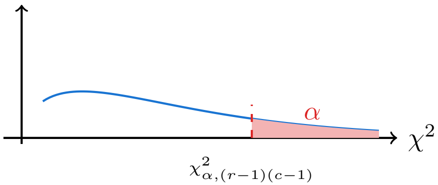
Reject H_0 if

$$\chi^{2*} > \chi^2_{\alpha,(r-1)(c-1)}$$

p-Value Method

Reject H_0 if

$$p\text{-value} < \alpha$$



Condition for a Valid Chi-Square Test

Expected Count Condition

The Chi-Square approximation is valid when:

- No more than **20%** of cells have expected count < 5
- Ideally, **all** expected counts ≥ 5

If the Condition Fails

- Combine categories where sensible
- Collect more data
- Use **Fisher's Exact Test**

Do Not Force Quantitative Data into Categories

If the response is quantitative, use ANOVA or regression – do not artificially bin the data.

Example 2: Political Affiliation vs. Opinion

Question: A random sample of 500 U.S. adults is questioned regarding their political affiliation and opinion on a tax reform bill. The results of this survey are summarized in the following contingency table:

	Favor	**Indifferent**	**Opposed**	**Total**
Democrat	138	83	64	285
Republican	64	67	84	215
Total	202	150	148	500

Identifying Variables:

- **Explanatory variable:** Party Affiliation (2 levels) $\Rightarrow$ rows
- **Response variable:** Opinion (3 levels) $\Rightarrow$ columns
- **Table size:** 2×3 (NOT 3×4)

Example 2: Expected Counts

	Favor	**Indifferent**	**Opposed**	**Total**
Democrat	$\frac{285 \times 202}{500} = 115.14$	$\frac{285 \times 150}{500} = 85.50$	$\frac{285 \times 148}{500} = 84.36$	285
Republican	$\frac{215 \times 202}{500} = 86.86$	$\frac{215 \times 150}{500} = 64.50$	$\frac{215 \times 148}{500} = 63.64$	215
Total	202	150	148	500

Intuition: Under H_0, 40.4% of *both* Democrats and Republicans should favour the bill. The expected table forces identical proportions across all rows.

Example 2: Computing the Statistic

	Favor	**Indifferent**	**Opposed**	**Total**
Democrat	138 (115.14)	83 (85.50)	64 (84.36)	285
Republican	64 (86.86)	67 (64.50)	84 (63.64)	215
Total	202	150	148	500

$$\chi^{2*} = 22.152, \qquad df = (2-1)(3-1) = 2$$

$$p\text{-value} < 0.001$$

Example 2: Conclusion

- Test statistic: $\chi^{2*} = 22.152$
- Degrees of freedom: $df = 2$
- Critical value at $\alpha = 0.05$: $\chi^2_{0.05,2} = 5.991$

Decision

Since $22.152 > 5.991$, we **reject** H_0.

Conclusion in Context

There is sufficient evidence that political affiliation and opinion on the tax reform bill are **associated**.

Intuition: Democrats favoured the bill more than expected under independence; Republicans opposed it more than expected.

Chi-Square Workflow I: Goodness-of-Fit Test

Procedure

1. Verify: random sample; all $E \geq 5$
2. State H_0: population follows the claimed distribution
3. Record observed counts O_i for each of k categories
4. Compute expected counts: $E_i = n\, p_{i,0}$
5. Verify all $E_i \geq 5$
6. Compute $\chi^{2*} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}, \quad df = k - 1$
7. Find p-value from χ^2_{k-1} distribution
8. Compare to α; conclude in context

Key Facts

- **One** categorical variable with $k \geq 2$ levels
- H_0 specifies *all* k probabilities $p_{1,0}, \ldots, p_{k,0}$
- $\sum p_{i,0} = 1$ must hold exactly
- $df = k - 1$ (one df lost estimating n)
- Test is always **right-tailed**
- Expected count: $E_i = n\, p_{i,0}$
- Large $\chi^{2*} \Rightarrow$ data deviate from claimed distribution

Quick Checklist

- ✓ All $E_i = np_{i,0} \geq 5$?
- ✓ $\sum p_{i,0} = 1$ exactly?
- ✓ $df = k - 1$, not k?
- ✓ Conclusion references the *distribution*, not a single category?

Chi-Square Workflow II: Test for Independence

Procedure

1. Verify conditions (random sample, independent observations)
2. State H_0 (independent) and H_a (dependent)
3. Build the observed contingency table
4. Compute $E = \dfrac{(\text{row total})(\text{col total})}{n}$
5. Check: all $E \geq 5$ (or $< 20\%$ cells below 5)
6. Compute $\chi^{2*} = \sum \dfrac{(O-E)^2}{E}$, $df = (r-1)(c-1)$
7. Find p-value from $\chi^2_{(r-1)(c-1)}$ distribution
8. Compare to α; state conclusion in context

Quick Checklist

- ✓ All $E \geq 5$?
- ✓ df uses $(r-1)(c-1)$, not rc?
- ✓ Conclusion says "association", not "cause"?
- ✓ p-value compared to α, not to 0.05 blindly?

Common Mistakes

- ! Using margin totals as cells in χ^2 sum
- ! Skipping the expected-count condition ($E < 5$ invalidates the approximation)
- ! Concluding *causation* from a statistically significant association
- ! Artificially binning quantitative data just to apply chi-square—use ANOVA instead
- ! Confusing table *size* ($r \times c$, excluding margins) with the grand total n

ANOVA: What It Is and When to Use It

What is ANOVA?

Analysis of Variance (ANOVA) tests whether the means of **three or more groups** are all equal.

When to Use It

- Response variable: **quantitative**
- Explanatory variable: **categorical** (the "factor")
- Factor has **more than two levels**

Learning goals:

- Understand the logic of ANOVA (between vs. within variation)
- Perform one-way ANOVA and read the ANOVA table
- Interpret F-statistic, p-value, and R^2
- Apply Tukey's multiple comparison procedure

ANOVA Hypotheses

Null Hypothesis

$$H_0: \; \mu_1 = \mu_2 = \cdots = \mu_t$$

All group (population) means are equal.

Alternative Hypothesis

$$H_a: \; \text{at least one mean is different from the others}$$

Important

H_a does **not** require *all* means to differ. Even one outlier mean is enough to reject H_0.

One-Way ANOVA: Main Question

Purpose

One-way ANOVA is used to compare the means of three or more groups.

Research Question

Are the population means of all groups equal, or is at least one mean different?

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_t$$

$$H_a : \text{at least one population mean is different}$$

Key Idea

ANOVA compares variation between group means with variation within the groups.

ANOVA Assumptions

1. **Normality:** each group population is approximately normal.
2. **Equal variances:** all groups have the same population variance.
3. **Independence:** observations are independent within and across groups.

Rule of Thumb for Equal Variances

If

$$\frac{\text{largest sample standard deviation}}{\text{smallest sample standard deviation}} < 2,$$

the equal-variance assumption is usually reasonable.

Remark

ANOVA is reasonably robust to mild violations, especially when group sizes are equal.

ANOVA Sums of Squares

Let t be the number of groups, n_i be the sample size in group i, and $n_T = \sum_{i=1}^{t} n_i$ be the total sample size.

Between Sum of Squares:

$$\text{SSB} = \sum_{i=1}^{t} n_i \left(\bar{y}_{i.} - \bar{y}\right)^2$$

Within-Samples Sums of Squares:

$$\text{SSW} = \sum_{i=1}^{t} \sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_{i.}\right)^2$$

Total Sum of Squares:

$$\text{TSS} = \sum_{i=1}^{t} \sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}\right)^2$$

Fundamental Identity

$$\text{TSS} = \text{SSB} + \text{SSW}$$

Shortcut Formulas for ANOVA Sums of Squares

Suppose there are t groups. Let

$$T_i = \text{sum of observations in group } i, \qquad n_i = \text{sample size in group } i.$$

Let n be the total number of observations.

Between-Samples Sum of Squares

$$\text{SSB} = \left(\frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \cdots + \frac{T_t^2}{n_t}\right) - \frac{(\sum y_i)^2}{n}$$

Within-Samples Sum of Squares

$$\text{SSW} = \sum y_i^2 - \left(\frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \cdots + \frac{T_t^2}{n_t}\right)$$

The ANOVA Table

Source	**df**	**SS**	**MS**	**F**
Treatment	$t-1$	SSB	$\text{MSB} = \dfrac{\text{SSB}}{t-1}$	$F = \dfrac{\text{MSB}}{\text{MSW}}$
Error	$n_T - t$	SSW	$\text{MSW} = \dfrac{\text{SSW}}{n_T - t}$	
Total	$n_T - 1$	TSS		

Decision Rule

If p-value $< \alpha$, reject H_0. There is evidence that at least one group mean is different.

Example: Pig Weights Under Four Diets

Problem

Twenty piglets are randomly assigned to four diet groups, with five piglets in each group. After 10 months, their weights are recorded. Test whether the mean weights differ across diets at $\alpha = 0.05$.

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

H_a : at least one mean weight is different

Obs.	**Feed 1**	**Feed 2**	**Feed 3**	**Feed 4**
1	60.8	68.3	102.6	87.9
2	57.1	67.7	102.2	84.7
3	65.0	74.0	100.5	83.2
4	58.7	66.3	97.5	85.8
5	61.8	69.9	98.9	90.3

Example: Pig Weights Under Four Diets

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$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_a : \text{at least one mean weight is different}$$

Obs.	Feed 1	Feed 2	Feed 3	Feed 4
1	60.8	68.3	102.6	87.9
2	57.1	67.7	102.2	84.7
3	65.0	74.0	100.5	83.2
4	58.7	66.3	97.5	85.8
5	61.8	69.9	98.9	90.3

Example: Group Totals

Feed	**Group Total**	n_i	**Mean**
Feed 1	303.4	5	60.68
Feed 2	346.2	5	69.24
Feed 3	501.7	5	100.34
Feed 4	431.9	5	86.38

Therefore,

$$\sum y = 303.4 + 346.2 + 501.7 + 431.9 = 1583.2$$

and

$$n = 20.$$

Also,

$$\sum y^2 = 130150.640.$$

Example: Compute SSB

First compute the grouped-square term:

$$\begin{aligned}\sum \frac{T_i^2}{n_i} &= \frac{303.4^2}{5} + \frac{346.2^2}{5} + \frac{501.7^2}{5} + \frac{431.9^2}{5} \\ &= 18409.156 + 23970.888 + 50340.578 + 37308.678 \\ &= 130029.300.\end{aligned}$$

The correction term is

$$\frac{(\sum y)^2}{n} = \frac{1583.2^2}{20} = 125326.112.$$

Therefore,

$$\begin{aligned}\text{SSB} &= 130029.300 - 125326.112 \\ &= 4703.188.\end{aligned}$$

Example: Compute SSW

The within-samples sum of squares is

$$\mathsf{SSW} = \sum y^2 - \sum \frac{T_i^2}{n_i}.$$

Substituting the values,

$$\begin{aligned}\mathsf{SSW} &= 130150.640 - 130029.300\\ &= 121.340.\end{aligned}$$

The total sum of squares is

$$\begin{aligned}\mathsf{SSTotal} &= \mathsf{SSB} + \mathsf{SSW}\\ &= 4703.188 + 121.340\\ &= 4824.528.\end{aligned}$$

Example: Mean Squares and Test Statistic

Here,

$$t = 4, \qquad n = 20.$$

The degrees of freedom are

$$df_{\text{between}} = t - 1 = 4 - 1 = 3,$$

and

$$df_{\text{within}} = n - t = 20 - 4 = 16.$$

Therefore,

$$\text{MSB} = \frac{\text{SSB}}{df_{\text{between}}} = \frac{4703.188}{3} = 1567.729,$$

and

$$\text{MSW} = \frac{\text{SSW}}{df_{\text{within}}} = \frac{121.340}{16} = 7.584.$$

Thus,

$$F = \frac{\text{MSB}}{\text{MSW}} = \frac{1567.729}{7.584} = 206.72.$$

Example: Complete ANOVA Table

Source	df	SS	MS	F
Between samples	3	4703.188	1567.729	206.72
Within samples	16	121.340	7.584	
Total	19	4824.528		

Conclusion

Using software or an F table, $p < 0.001$. Since $p < 0.001 < 0.05$, we reject H_0. There is strong evidence that the mean pig weights are not all equal across the four diets.

After ANOVA: Tukey's HSD

Why Do We Need Tukey's Test?

ANOVA tells us that at least one group mean is different. It does not tell us which specific group means differ.

Feed	n	**Mean (kg)**	**Tukey Grouping**
Feed 3	5	100.34	A
Feed 4	5	86.38	B
Feed 2	5	69.24	C
Feed 1	5	60.68	D

Different letters mean significantly different group means. Therefore, all four diet means are significantly different.

ANOVA in R

```
model <- aov(weight ~ feed, data = pigdata)

summary(model)

TukeyHSD(model)
```

Data Format

The data should be in long format: one column for weight and one column for feed group.

Key Takeaways

1. One-way ANOVA compares the means of three or more groups.
2. SSB measures variation between the group totals or means.
3. SSW measures variation within the groups.
4. The test statistic is $F = \text{MSB}/\text{MSW}$.
5. A large F value gives evidence against H_0.
6. If ANOVA is significant, Tukey's HSD can identify which pairs of means differ.